

Separation, Pooling, and Endogenous Timing

A Tullock Contest under Two-sided Asymmetry

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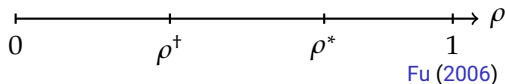
Master's Thesis Defense

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Research Question

- **Fu (2006)**: in a Tullock contest with private information, the informed player prefers to move *second* – revealing type early is costly.
- But this rests on $\rho = 1$: prize values are *perfectly correlated*.
- **Gap**: does the late-mover preference survive at $\rho < 1$, or is it an artifact of full correlation?
- **This paper**: introduce $\rho \in [0, 1]$ – at $\rho = 0$ the signal channel is closed; at $\rho = 1$ Fu's result is recovered as the limiting case.



What This Paper Does

- ***i.i.d.* benchmark** ($\rho = 0$) – isolates timing from information spillover.
- **Partial correlation** ($0 < \rho < 1$) as the main model – $\rho = 1$ is the endpoint, not a separate model.
- Characterizes undistorted vs. Riley distorted separation, and their transition as ρ rises.
- Links subgame equilibria to **endogenous timing**: a structural asymmetry between I and U .

Related Literature

Strand	Key reference	Main takeaway
Contest success function	Tullock (1980)	Ratio-form CSF; unique equilibrium at $r = 1$
Exogenous timing	Dixit (1987)	Stronger player overcommits first (top-dog)
Endogenous timing	Baik and Shogren (1992)	Weaker player moves first
Timing framework	Hamilton and Slutsky (1990)	SS / UI / IU timing architecture
Asymmetric info (SS)	Hurley and Shogren (1998)	Type uncertainty suppresses U 's effort
Information advantage	Wärneryd (2003)	Informed player can win less

Gap and Contribution

Strand	Key reference	Main takeaway
Signaling in contests	Fu (2006)	$\rho = 1$; late mover avoids revelation cost
Least-cost separation	Riley (1979)	Least-cost separation under binding IC
Timing reversal	Protopappas (2023, 2025)	Early-mover equilibrium under cost asymmetry
Gap	—	<i>i.i.d.</i> $\rightarrow$ fully correlated transition uncharacterized
This paper	—	One ρ spans the spectrum; re-opens the signal channel

Model Setup

- Two players: informed (I) and uninformed (U)
- Types: $V_I, V_U \in \{v_H, v_L\}$, $\Pr(V_k = v_H) = q \in (0, 1)$
- **Benchmark assumption:** $V_I \perp V_U$ (*i.i.d.*, $\rho = 0$)

	$V_U = v_H$	$V_U = v_L$
$V_I = v_H$	q^2	$q(1 - q)$
$V_I = v_L$	$q(1 - q)$	$(1 - q)^2$

- Three-stage timing: Stage 0 (choose timing) $\rightarrow$ Nature draws types $\rightarrow$ effort subgame
- Three regimes: SS (simultaneous), UI (U moves first), IU (I moves first)

The Closed Signal Channel

Lemma (Conditional Expectation Invariance)

If $V_I \perp V_U$, then

$$\mathbb{E}[V_U | x_I] = \mathbb{E}[V_U], \quad \forall x_I.$$

→ **Signal channel closed:** beliefs update via Bayes, but U 's best response is unchanged – *updating happens, but it doesn't bite.*

Proposition (Unique Separating Equilibrium)

The IU subgame admits a *unique separating equilibrium*, with IC slack

$$\frac{(v_H - v_L)^2}{4\bar{v}} > 0, \quad \bar{v} \equiv \mathbb{E}[V_k].$$

→ I 's lead reveals her type at *no cost*: the IC constraint never binds.

Equilibrium Payoffs across Regimes

Composite coefficients: $\hat{A} \equiv \bar{v} \mathbb{E}[V_I^{-1/2}]$, $\hat{B} \equiv \bar{v} \mathbb{E}[V_I^{-1}] \geq 1$ (Jensen).

Regime	x_U^*	$\tilde{\pi}_I$	$\tilde{\pi}_U$
SS	$\frac{\hat{A}^2}{(1 + \hat{B})^2}$	$\mathbb{E}\left[\left(\sqrt{V_I} - \frac{\hat{A}}{1 + \hat{B}}\right)^2\right]$	$\frac{\hat{A}^2 \hat{B}}{(1 + \hat{B})^2}$
UI	$\frac{\hat{A}^2}{4}$	$\mathbb{E}\left[\left(\sqrt{V_I} - \frac{\hat{A}}{2}\right)^2\right]$	$\frac{\hat{A}^2}{4}$
IU (sep.)	$\frac{v(2\bar{v} - v)}{4\bar{v}}$	$\frac{\mathbb{E}[V_I^2]}{4\bar{v}}$	$\frac{\mathbb{E}[(2\bar{v} - V_I)^2]}{4\bar{v}}$

→ **First-mover advantage for U:** $\tilde{\pi}_U^{UI} - \tilde{\pi}_U^{SS} = \frac{\hat{A}^2(1 - \hat{B})^2}{4(1 + \hat{B})^2} \geq 0$.

Stage-0 Timing Game

Four unilateral deviation gaps:

$$\Delta_I^{UI} \equiv \tilde{\pi}_I^{UI} - \tilde{\pi}_I^{SS}, \quad \Delta_I^{IU} \equiv \tilde{\pi}_I^{IU} - \tilde{\pi}_I^{SS}, \quad \Delta_U^{UI} \equiv \tilde{\pi}_U^{UI} - \tilde{\pi}_U^{SS}, \quad \Delta_U^{SS} \equiv \tilde{\pi}_U^{SS} - \tilde{\pi}_U^{IU}$$

Proposition (Stage-0 Pure-strategy Equilibria)

Pure-strategy Nash equilibria are fully characterized by the signs of the four gaps:

- UI iff $\Delta_I^{UI} \geq 0$
- SS iff $\Delta_I^{UI} \leq 0$ and $\Delta_U^{SS} \geq 0$
- IU iff $\Delta_I^{IU} \geq 0$ and $\Delta_U^{SS} \leq 0$

→ **Takeaway:** timing is driven purely by gap signs – no signaling distortion.
→ Any effect at $\rho > 0$ must come from the reopened channel – not from asymmetric information per se.

Correlated Type Structure

- Pearson correlation: $\rho \equiv \text{corr}(H_I, H_U)$, where $H_k = \mathbf{1}\{V_k = v_H\}$, $\rho \in [0, 1]$
- Marginals fixed at $(q, 1 - q)$ – ρ is the single free parameter of the joint distribution

	$V_U = v_H$	$V_U = v_L$
$V_I = v_H$	$q^2 + \rho q(1 - q)$	$q(1 - q)(1 - \rho)$
$V_I = v_L$	$q(1 - q)(1 - \rho)$	$(1 - q)^2 + \rho q(1 - q)$

ρ	Structure	Key property
$\rho = 0$	<i>i.i.d.</i> benchmark	signal has no spillover
$0 < \rho < 1$	partial correlation	belief enters U 's best response
$\rho = 1$	fully correlated	Fu (2006) endpoint

Posterior Expected Valuation

Conditional probabilities of U 's high state, given I 's type:

$$\theta_H = q + \rho(1 - q), \quad \theta_L = q(1 - \rho).$$

Lemma (Posterior Expected Valuation)

Given posterior belief $\mu = \Pr(V_I = v_H \mid x_I)$,

$$\tilde{V}_U(\mu) = v_L + [q + \rho(\mu - q)]\Delta V, \quad \frac{d\tilde{V}_U(\mu)}{d\mu} = \rho\Delta V > 0.$$

Posterior μ	Interpretation	$\tilde{V}_U(\mu)$
$\mu = 1$	I is high type	$v_L + [q + \rho(1 - q)]\Delta V$
$\mu = q$	beliefs unchanged	$v_L + q\Delta V = \mathbb{E}[V_U]$
$\mu = 0$	I is low type	$v_L + q(1 - \rho)\Delta V$

→ When $\rho > 0$, I 's type is a signal about U 's own prize — the channel is live.

SS and UI under Partial Correlation

Composite coefficients – replace $\bar{v}$ with $\tilde{V}_U(1), \tilde{V}_U(0)$:

$$\tilde{A} \equiv q \frac{\tilde{V}_U(1)}{\sqrt{v_H}} + (1 - q) \frac{\tilde{V}_U(0)}{\sqrt{v_L}}, \quad \tilde{B} \equiv q \frac{\tilde{V}_U(1)}{v_H} + (1 - q) \frac{\tilde{V}_U(0)}{v_L}.$$

- Equilibrium structure identical in form to the benchmark: $\hat{A}, \hat{B} \rightarrow \tilde{A}, \tilde{B}$ (at $\rho = 0$: $\tilde{A} = \hat{A}, \tilde{B} = \hat{B}$)

Proposition (First-mover Advantage Robust)

$\tilde{\pi}_U^{UI} \geq \tilde{\pi}_U^{SS}$ for all $\rho \in [0, 1)$.

- **Monotone coefficients:** $\tilde{A}, \tilde{B}$ strictly decreasing in ρ , with $\tilde{B} \rightarrow 1$ as $\rho \rightarrow 1$ – rising ρ erases the Jensen excess burden $\hat{B} > 1$

When Does Separation Hold?

Undistorted separating candidate in the IU subgame, and its IC condition:

$$x_H = \frac{v_H^2}{4\tilde{V}_U(1)}, \quad x_L = \frac{v_L^2}{4\tilde{V}_U(0)}; \quad \frac{\tilde{V}_U(1)}{\tilde{V}_U(0)} \leq R \equiv \frac{v_H^2}{v_L(2v_H - v_L)}.$$

Proposition (Separation Threshold)

Undistorted separation holds iff $\rho \leq \rho^*$, where

$$\rho^* = \frac{(R - 1) \tilde{V}_U(q)}{(1 - q + Rq) \Delta V}.$$

- At $\rho = 0$: strictly satisfied; at $\rho = 1$: violated
- IC slack $S(\rho) \equiv R - \tilde{V}_U(1)/\tilde{V}_U(0)$ decreases monotonically and hits 0 exactly at ρ^*

Riley Distortion

When $\rho > \rho^*$, define

$$\alpha \equiv 1 - \frac{\tilde{V}_U(0)}{\tilde{V}_U(1)} \in (0, 1), \quad z_H^- \equiv \frac{v_H}{2\sqrt{\tilde{V}_U(0)}}(1 - \sqrt{\alpha}).$$

- High type keeps undistorted effort x_H ; low type distorts *down* to $x_L^R = (z_H^-)^2$

Corollary (Two-region Structure)

- $\rho \leq \rho^*$: undistorted separation – no signaling cost
- $\rho > \rho^*$: Riley distorted separation – separation cost borne by the low type alone

→ As ρ rises, posterior prize-value dispersion widens; the original low-type effort no longer supports separation, so the low type must distort downward to keep deviations unprofitable.

Stage-0 Payoff Matrix

Key point: at stage 0, types are unrealized — players compare *ex ante* expected payoffs.

	U : Early	U : Late
I : Early	$(\tilde{\pi}_I^{SS}, \tilde{\pi}_U^{SS})$	$(\tilde{\pi}_I^{IU}, \tilde{\pi}_U^{IU})$
I : Late	$(\tilde{\pi}_I^{UI}, \tilde{\pi}_U^{UI})$	$(\tilde{\pi}_I^{SS}, \tilde{\pi}_U^{SS})$

Proposition (Stage-0 Pure-strategy Equilibria)

Same sign characterization as the benchmark $(\Delta_I^{UI}, \Delta_I^{IU}, \Delta_U^{UI}, \Delta_U^{SS})$ — but now *every entry depends on ρ* .

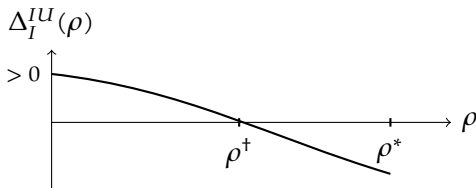
→ The question becomes: **how do the gap signs move with ρ ?**

The Structural Asymmetry

- **U 's side:** $\Delta_U^{IU} \geq 0$ for all $\rho < 1$ – U always wants to lead against a late-moving I (robust)
- **I 's side:** $\tilde{\pi}_I^{IU}$ strictly decreasing in ρ – revelation cost rises with correlation

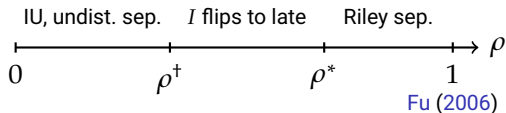
Proposition (Timing Reversal)

If $\Delta_I^{IU}(0) > 0$ and $\Delta_I^{IU}(\rho^*) < 0$, there exists $\rho^\dagger \in (0, \rho^*)$ where I 's timing preference flips from early to late.



→ The reversal is driven entirely by I , not U .

Summary – The Correlation Spectrum



ρ range	IU equilibrium type	I 's timing preference
$[0, \rho^+)$	undistorted separation	lead (IU)
$(\rho^+, \rho^*]$	undistorted separation	wait
$(\rho^*, 1)$	Riley distorted separation	wait
$\rho = 1$	Fu (2006) endpoint	wait (late mover)

→ **Bottom line:** **Fu (2006)**'s late-mover result *requires* correlation – it is the right endpoint, not a general principle.

Welfare Framework

Social welfare = allocative value minus effort dissipation: $\widetilde{W} = \widetilde{G} - \widetilde{T}$, where $\widetilde{G} = \mathbb{E}[p_I V_I + p_U V_U]$ and $\widetilde{T} = \mathbb{E}[x_I + x_U]$.

Equilibrium type	Expected dissipation $\widetilde{T}$	Sensitivity to ρ
SS	$\frac{\widetilde{A}}{1 + \widetilde{B}} \mathbb{E}[\sqrt{V_I}]$	ambiguous ($\widetilde{A}, \widetilde{B}$ both fall)
UI	$\frac{\widetilde{A}}{2} \mathbb{E}[\sqrt{V_I}]$	strictly decreasing (via $\widetilde{A}$)
IU undistorted sep.	$\frac{\mathbb{E}[V_I]}{2}$	immune to ρ
IU Riley distorted	$\frac{v_H}{2} [1 - (1 - q)\sqrt{\alpha}]$	varies via α
IU pooling	$\sqrt{x_P \bar{v}}$	no direct ρ term for fixed x_P

→ **IU undistorted separation's dissipation is insulated from ρ** – the key structural difference.

Three Effects of ρ on Welfare

- **Allocative effect:** $\rho \uparrow \rightarrow \tilde{V}_U(1)$ and $\tilde{V}_U(0)$ diverge $\rightarrow$ win probabilities and prize allocation shift across paths; the welfare effect is not sign-definite a priori
- **Dissipation effect:** regime-dependent — UI dissipation falls with ρ (via $\tilde{A}$), SS is ambiguous; only IU undistorted is *insulated* at $\mathbb{E}[V_I]/2$
- **Threshold effect:** ρ crosses ρ^* $\rightarrow$ Riley distortion begins $\rightarrow$ the low-type path is distorted, but welfare need not jump discontinuously at ρ^*

Key implication

- In IU undistorted separation, $\tilde{W}^{\text{sep}}$ decreases with ρ even though dissipation stays fixed
- Across regimes, welfare rankings remain parameter-dependent: SS and UI react through $\tilde{A}$, $\tilde{B}$, while Riley welfare must be judged from the full formula

Equilibrium vs. Social Optimum

- **Direction 1** ($\rho > \rho^+$): equilibrium is SS, yet $\tilde{W}^{\text{sep}} > \tilde{W}^{\text{SS}}$ – *I's retreat sustains the welfare-inferior regime*
- **Direction 2** (IU remains privately sustainable): equilibrium may stay in IU Riley or pooling, yet $\tilde{W}^{\text{SS}}$ or $\tilde{W}^{\text{UI}}$ can still be higher – *private timing incentives and social welfare need not align*

Root cause

I does not internalize *U*'s payoff; private timing incentives ignore the global dissipation–allocation trade-off. Efficiency of equilibrium timing is *not* a general result – it depends on the parameter configuration.

Welfare across Regimes

- **IU undistorted separation:** $\tilde{T}^{\text{sep}} = \mathbb{E}[V_I]/2$, while $\tilde{W}^{\text{sep}}$ decreases with ρ
- **SS and UI:** both move with $\tilde{A}, \tilde{B}$; neither ordering is sign-definite
- **Riley region:** both dissipation and allocation change, so welfare must be read from the full formula
- **At $\rho = \rho^*$:** the two expressions meet; regime change need not imply a welfare jump

Takeaway

Welfare rankings are regime- and parameter-dependent.

Conclusion

- 1 $\rho = 0$: unique undistorted separation — I leads at *no revelation cost*.
- 2 $\rho > 0$ reopens the signal–belief–response channel: beliefs enter U 's best response.
- 3 **Separation threshold**: undistorted iff $\rho \leq \rho^*$; beyond, Riley distortion — cost borne by the low type alone.
- 4 **Timing asymmetry**: U 's lead preference is robust; I flips to late at $\rho^+ < \rho^*$.
- 5 **Fu (2006)** is the $\rho = 1$ endpoint — not a general principle.
- 6 **Welfare**: equilibrium timing can depart from the social optimum; welfare responses to ρ are regime-dependent.

Future Directions

- ① **Continuous type space** – beyond the closed-form threshold: richer distortion patterns.
- ② **Negative correlation** ($\rho < 0$) – reverse information spillover.
- ③ **N -player contests** – timing symmetry-breaking; the role of correlation remains open.